A Comprehensive Review of Agricultural Parcel and Boundary Delineation from Remote Sensing Images: Recent Progress and Future Perspectives

Juepeng Zheng et al.

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Chapter 1

Research Summary

Preface

This report provides a comprehensive summary and critical analysis of *A Comprehensive Review of Agricultural Parcel and Boundary Delineation from Remote Sensing Images: Recent Progress and Future Perspectives* by Zheng et al. The following sections synthesize the authors' framework, methodology, and key findings regarding Agricultural Parcel and Boundary Delineation (APBD)

1.1 Introduction

Definition 1.1.1: Agricultural Parcel and Boundary Delineation

Agricultural Parcels represent the fundamental spatial units for agricultural activity.

Agricultural Parcel and Boundary Delineation (APBD) is an interdisciplinary application of computer algorithms for vision tasks where agricultural fields are disected into their component parcels, and distinctly separated from their background (adjacent parcels, natural vegetation and other types of land).

Applications of APBD include:

- sustainable agriculture planning and land-use monitoring
- crop mapping
- yield estimation

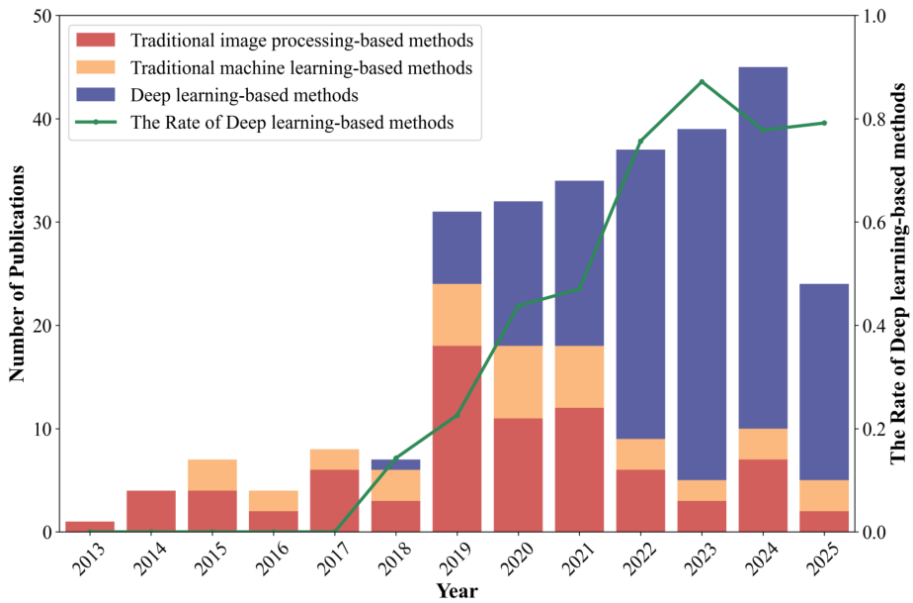
Foundation models such as Transformer architectures, the Segment Anything Model (SAM), and Vision-Language Vision-Language Models (VLMs), leveraging massive datasets and self-supervised learning, offer strong generalizaton ability, high adaptability, and improved performance accross diverse agricultural scenarios when applied to high-resolution remote sensing imagery.

The authors conceptualize APBD as composing three hierarchical levels:

1. Cropland Identification
2. Boundary Delineation
3. Parcel Segmentation

The paper seeks to synthesize the emerging trends in APBD research, presenting the historical evolution, current status, and future directions. The authors identify remote sensing data as a critical driver of APBD research in the foreseeable future.

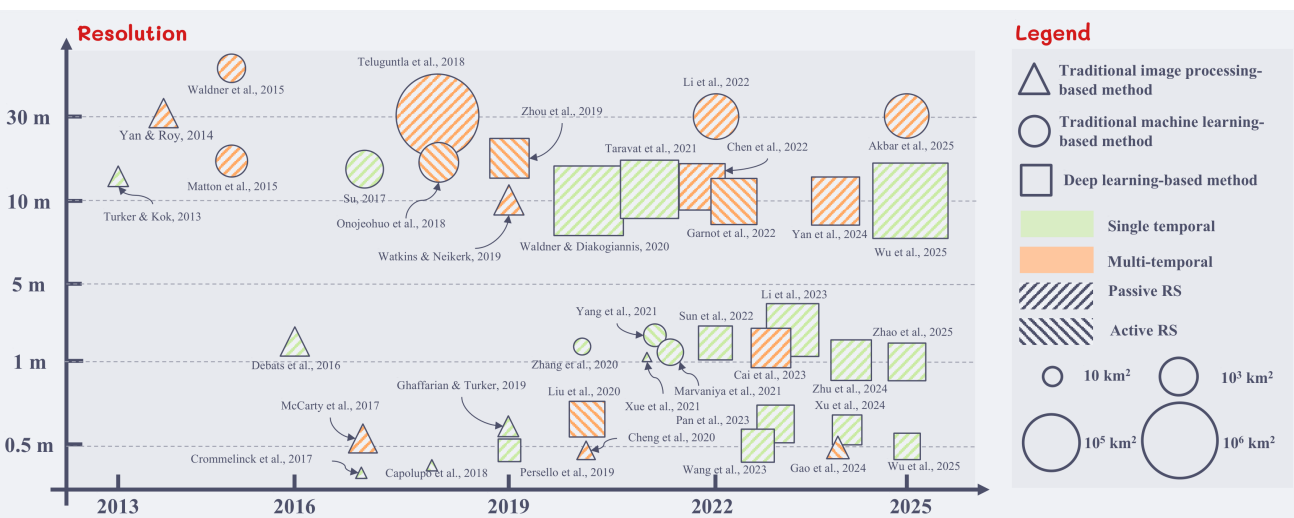
1.2 Meta-analysis of Related Literature



Trends in APBD methodologies (Figure 2)

I Overall Trend

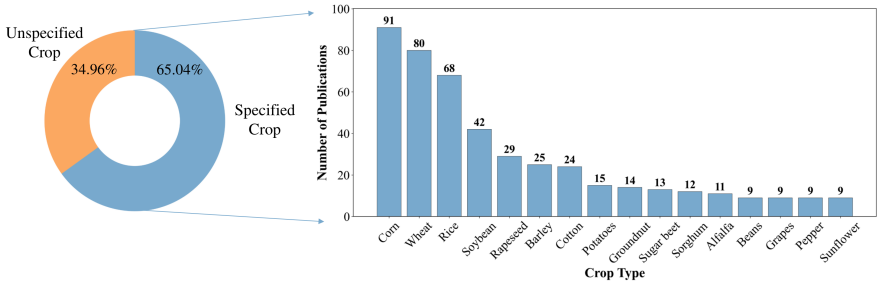
- *"Since 2019, the number of APBD studies using Very High-Resolution (VHR) imagery has increased exponentially."*
- Although VHR imagery enables high accuracy APBD, storage and processing costs prove to be a disadvantage. Large scale studies tend to use >10 m spatial resolution to balance land variety and coverage, accuracy, and storage.



Overall trends of APBD development (Figure 3)

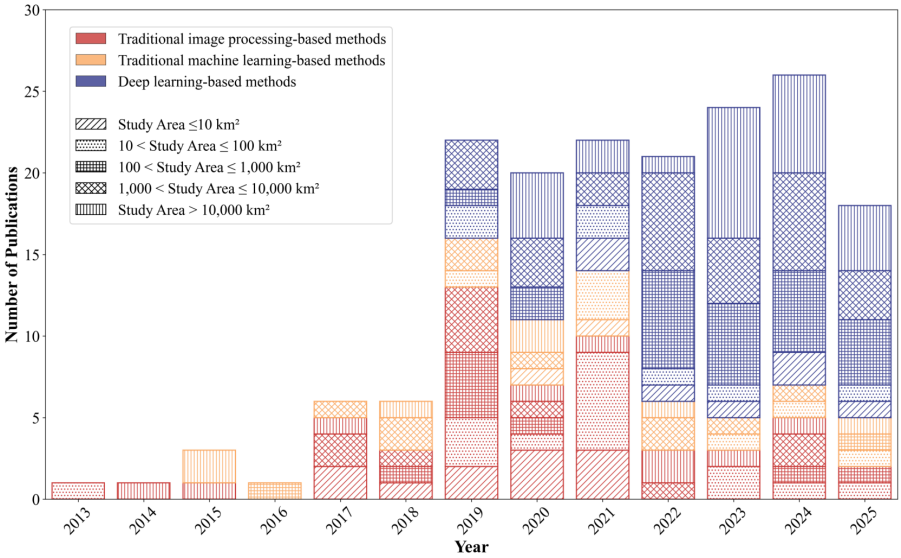
II Quantitative Analysis

- *Types of crops:* Corn crop most widely studied for APBD with statistics presented in figure 4 of the paper, attached below.



Statistics of crop types in APBD-related publications (Figure 4)

- *Study sites:* China (130 studies) leads in volume of publications followed by USA (19), Netherlands (14), and India (11).
- *Study area:* Rising trend in extremely large region studies (>10,000 km) catalysed by deep learning techniques.



Study area statistics for APBD publications (Figure 6)

- *Sensor type:* A strong reliance on satellite data (84.0%), followed by Unmanned Aerial Vehicles (10.8%) and aerial imagery (5.2%). Sentinel (82) and Landsat (31) lead the list in medium resolution satellite imagery, while GF series (62) is the overwhelming majority in high-resolution ($\leqslant$ 1m) studies.
- *Spatial resolution of data:* An increasing reliance on high-resolution imagery, attributed to the advent of modern deep learning algorithms that can extract detailed texture and spatial information.

III Methodology Review

Traditional image processing-based methods

Acknowledgment

This report is based entirely on the work of Zheng et al. All concepts, findings, and frameworks presented herein are derived from their original research. No additional sources have been consulted for the core content of this summary.

References

Zheng, Juepeng et al. (2025). *A Comprehensive Review of Agricultural Parcel and Boundary Delineation from Remote Sensing Images: Recent Progress and Future Perspectives*. arXiv: 2508.14558 [cs.CV]. URL: https://arxiv.org/abs/2508.14558.